

AI-BASED HANDWRITTEN ASSIGNMENT AND COPIED CONTENT DETECTION SYSTEM

MINI PROJECT REPORT

Name: _____

Roll No.: _____

Class/Semester: _____

Department: _____

College: _____

Project Guide: _____

Academic Year: 2026–2027

Certificate

This is to certify that the project entitled “**AI-Based Handwritten Assignment and Copied Content Detection System**” is a bonafide project work carried out by _____ under the guidance of _____ in partial fulfillment of the requirements for the completion of the course.

Project Guide

Head of Department

Date

Acknowledgement

We would like to express our sincere gratitude to our project guide, faculty members, and department for their valuable guidance and support throughout the development of this project.

We are also thankful to our college for providing the required resources and facilities for completing this project.

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Abstract

The **AI-Based Handwritten Assignment and Copied Content Detection System** is designed to detect similarities between handwritten assignments submitted by students. The main objective of the system is to reduce manual effort in checking assignments and help teachers identify potentially copied content.

The system accepts handwritten assignment images or scanned documents as input. Image processing techniques are applied to improve the quality of the input image. A handwriting Optical Character Recognition (OCR) model is then used to convert handwritten content into machine-readable text.

After extracting the text, the system compares it with previously submitted assignments stored in the database. Text similarity techniques such as **TF-IDF and Cosine Similarity** can be used to calculate the similarity percentage.

The system is intended to support teachers rather than automatically accuse students of plagiarism.

1. Introduction

In educational institutions, students frequently submit handwritten assignments. Teachers generally have to check these assignments manually to determine their correctness and whether students have copied answers from one another.

Manual identification of copied assignments is time-consuming, especially when the number of students is large. Handwritten assignments also cannot easily be compared using traditional text-based plagiarism detection systems because the content first needs to be converted into digital text.

The proposed system addresses this problem by combining handwriting recognition, image processing, and text similarity analysis.

2. Problem Statement

Teachers face difficulty in identifying copied content in handwritten assignments because:

- Assignments are submitted in handwritten form.
- Manual comparison requires significant time.
- Similar content may appear across many assignments.
- Traditional plagiarism tools generally require digital text.
- Teachers need a convenient way to review suspicious submissions.

Therefore, there is a need for an automated system that can convert handwritten assignments into text and identify similarities between submissions.

3. Objectives

1. Accept handwritten assignment images as input.
2. Preprocess assignment images for better recognition.
3. Convert handwritten text into digital text using OCR.
4. Store extracted assignment text in a database.
5. Compare assignments automatically.
6. Calculate a similarity percentage.
7. Identify potentially copied content.
8. Provide a simple teacher dashboard.
9. Generate a report containing similarity results.
10. Reduce manual effort required for assignment checking.

4. Existing System

In the existing system, teachers generally check handwritten assignments manually. The teacher reads each assignment and compares answers with other students' assignments or reference material.

Disadvantages

- Time-consuming.
- Requires considerable manual effort.
- Difficult to compare a large number of assignments.
- Human comparison may miss similarities.
- No automatic similarity percentage.
- Difficult to maintain historical assignment records.

5. Proposed System

The proposed system uses Artificial Intelligence and image processing to analyze handwritten assignments.

The student or teacher uploads an assignment image. The system preprocesses the image and extracts handwritten text using an OCR model.

The extracted text is stored in a database. When a new assignment is submitted, its text is compared with previously stored assignments.

Advantages

- Reduces manual work.
- Supports handwritten assignments.
- Automatically extracts text.
- Provides similarity percentage.
- Stores previous assignments.
- Makes suspicious assignments easier to review.
- Can be extended with semantic similarity models.

6. System Architecture

USER — Student / Teacher



Assignment Upload



Image Preprocessing —
OpenCV



Handwriting OCR — TrOCR



Extracted Text



Assignment Database



Similarity Analysis — TF-IDF /
Embeddings



Similarity Score + Matching
Content



Teacher Dashboard / Report

7. Working of the System

Step 1: Assignment Upload

The user uploads a handwritten assignment as an image or scanned document.

- JPG
- JPEG
- PNG
- PDF

Step 2: Image Preprocessing

OpenCV can be used for grayscale conversion, noise removal, thresholding, contrast improvement, deskewing, and cropping.

Step 3: Handwriting OCR

The preprocessed image is passed to a handwriting recognition model such as TrOCR.

Handwritten: "Photosynthesis is the process by which green plants prepare their fo



Step 4: Text Cleaning

OCR output is cleaned and normalized before comparison.

Step 5: Database Storage

Extracted text is stored along with student information, assignment details, and submission date.

Step 6: Similarity Detection

The new assignment is compared with existing assignments using TF-IDF and Cosine Similarity.

Step 7: Result Generation

Student: Rahul Assignment: Science Assignment 1 Similarity: 87% Status: POSSIBLE C

8. Modules

8.1 User Management Module

Manages students, teachers, administrators, and their respective roles.

8.2 Assignment Upload Module

Allows users to upload handwritten assignments in image or PDF format.

8.3 Image Processing Module

Improves assignment images before OCR using OpenCV.

8.4 Handwriting Recognition Module

Converts handwritten text into machine-readable text using a handwriting OCR model.

8.5 Similarity Detection Module

Compares extracted text with existing submissions.

8.6 Database Module

Stores student information, assignments, OCR text, and similarity results.

8.7 Teacher Dashboard Module

Displays assignments, similarity percentages, matching content, and reports.

8.8 Report Module

Generates a report containing student information, OCR text, similarity percentage, matching content, and final status.

9. Technologies Used

Component	Technology
Programming Language	Python
Frontend	HTML, CSS, JavaScript / Streamlit
Backend	Flask / FastAPI
Image Processing	OpenCV
OCR	TrOCR
Text Processing	Python NLP
Similarity	TF-IDF + Cosine Similarity
Database	MySQL
Machine Learning	PyTorch / Transformers

10. Database Design

Student Table

Student_ID Student_Name Email Class Department

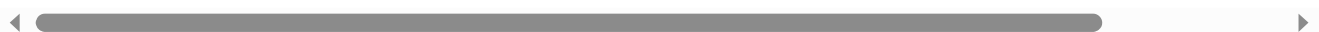
Assignment Table

Assignment_ID Student_ID Subject Assignment_Title File_Name Submission_Date OCR_Te



Similarity Table

Similarity_ID Assignment_ID Compared_Assignment_ID Similarity_Percentage Status An



Teacher Table

Teacher_ID Teacher_Name Email Password Department

11. Similarity Algorithm

TF-IDF is used to represent text numerically. TF-IDF stands for Term Frequency–Inverse Document Frequency.

After converting documents into vectors, Cosine Similarity is used to calculate how similar the documents are.

$$A \cdot B \text{ Similarity} = \frac{A \cdot B}{||A|| \times ||B||}$$

The resulting value can be converted into a percentage.

$$\text{Similarity} = 0.87 \quad \text{Similarity Percentage} = 87\%$$

Percentage	Interpretation
0–49%	Low Similarity
50–79%	Medium Similarity
80–100%	High Similarity

Similarity thresholds should be configurable. A high similarity score is an indicator for teacher review and should not automatically be treated as proof of plagiarism.

12. Example Result

Assignment A

Plants prepare their food through the process of photosynthesis using sunlight, wa



Assignment B

Green plants make their food by photosynthesis using sunlight, water and carbon di



OCR Confidence: 89% Text Similarity: 84% Status: POSSIBLE COPIED CONTENT Acti



13. Expected Output

===== ASSIGNMENT COPY REPORT =====



14. Advantages

- Saves teacher time.
- Automates handwritten text extraction.
- Detects similarities between assignments.
- Provides measurable similarity scores.
- Maintains assignment history.
- Helps teachers identify suspicious submissions.
- Can process a large number of assignments.
- Can be expanded with advanced AI models.

15. Limitations

1. Handwriting recognition may be inaccurate for very poor handwriting.
2. Blurred or low-quality images can reduce OCR accuracy.
3. Different handwriting styles can affect recognition.
4. Mathematical equations and diagrams may require specialized recognition.
5. OCR errors can affect similarity results.
6. Common textbook answers may naturally have high similarity.
7. Similarity detection should not be treated as automatic proof of plagiarism.

16. Future Scope

Better Handwriting Recognition

Advanced handwriting recognition models can improve OCR accuracy.

Semantic Similarity

Sentence Transformer models can detect similar meaning even when students use different words.

Online Source Checking

The system can be extended to compare extracted text against approved online or reference sources.

Multi-language Support

The system can be extended to support Hindi, English, and other languages.

Mobile Application

Teachers could photograph assignments directly using a mobile application.

Automatic Question-Answer Checking

The system could identify questions and answers and compare answers against a teacher-provided answer key.

AI-Based Feedback

AI could provide suggestions and explanations for detected similarities.

17. Testing

Test Case	Input	Expected Result
TC01	Clear handwriting	Correct OCR
TC02	Poor handwriting	Partial OCR
TC03	Two identical assignments	High similarity
TC04	Two different assignments	Low similarity
TC05	Similar wording	Medium/high similarity
TC06	Blank image	No text detected
TC07	Low-quality image	Warning / reduced OCR accuracy
TC08	Multiple submissions	Correct database comparison

18. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or equivalent
- RAM: 8 GB minimum
- Storage: 10 GB available
- Camera or scanner for assignment images
- Internet connection for downloading AI models if required

Recommended Hardware

- Intel Core i5/i7 or equivalent
- RAM: 16 GB
- GPU with CUDA support for faster OCR processing

Software Requirements

- Windows / Linux / macOS
- Python 3.x
- VS Code or PyCharm
- Git
- MySQL
- Required Python libraries

19. Conclusion

The **AI-Based Handwritten Assignment and Copied Content Detection System** provides an automated approach for analyzing handwritten assignments.

The system combines image preprocessing, handwriting OCR, text processing, and similarity analysis. It converts handwritten assignments into digital text and compares the extracted content with previously submitted assignments.

The proposed system can reduce the manual effort required by teachers and provide useful information for identifying potentially copied assignments.

Final decisions regarding copying or plagiarism should be made by the teacher after reviewing the original assignments and the similarity report.

20. References

1. Microsoft Research — TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models.
2. Hugging Face Transformers Documentation — TrOCR.
3. OpenCV Documentation — Image Processing and Computer Vision.
4. Scikit-learn Documentation — TF-IDF and Cosine Similarity.
5. Python Documentation — Python Programming Language.
6. PyTorch Documentation — Machine Learning Framework.

Project Summary

Item	Details
Project Title	AI-Based Handwritten Assignment and Copied Content Detection System
Domain	Artificial Intelligence / Machine Learning / NLP
Input	Handwritten Assignment Image / PDF
Processing	Image Processing + Handwriting OCR + Text Similarity
Output	Extracted Text + Similarity Percentage + Copy Detection Report
Primary Users	Teachers and Educational Institutions
Main Benefit	Reduces manual effort and assists teachers in identifying potentially copied handwritten assignments.

AI-Based Handwritten Assignment and Copied Content Detection System

Mini Project Report — Academic Year 2026–2027